

VICTORIA JI

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TECHNICAL SKILLS

Firmware & Embedded: C++, C, Embedded Systems, Arduino Microcontrollers, Low-Level Embedded Development, Hardware/Software Integration, Sensor Integration, Signal Processing, Low-Power Design

Debugging & Validation: Hardware Diagnostics, Software Debugging, Fault Isolation, Troubleshooting, Iterative Testing, Failure Analysis, Unit Testing, Test Coverage

AI & Software: Python, NLP, RAG Systems, LLM Fine-Tuning, Data Processing, Web Scraping, Android Development, Java, C#, JavaScript, TypeScript

Development Tools: Git, Version Control, React, Next.js, Node.js, Tailwind CSS, L^AT_EX, Web Visualization, UI/UX

Languages: English, Mandarin, Cantonese

EXPERIENCE

Senior Software Engineering Intern | Appli AI

Oct 2025 – Present

Promoted from: Software Engineering Intern | Appli AI

Jun 2025 – Oct 2025

- Contribute to **Appli AI**, an **LOI Batch 15 startup** automating hiring workflows through **AI-powered algorithms**, with work spanning NLP, RAG systems, software development, research, and product execution.
- Independently developed a **LinkedIn profile-matching algorithm** that accepts a description, processes it using NLP, scrapes LinkedIn profile data, and ranks matching candidates by relevance to help recruiters identify targeted talent.
- Independently built an **NLP-powered FAQ generator** that converts large text inputs into potential frequently asked questions and corresponding answers.
- Contributed to a **RAG vector database**; collected and analyzed **50+ statistics** on predictors of job performance and their correlation indices to support product and model research.
- Researched **RAG-based model fine-tuning** and wrote technical reports documenting findings, implementation considerations, and potential improvements.
- Supported company growth by connecting Appli AI with funded startups through social platforms and personal networks, while engaging founders and developers across Canada to study trends in **B2B SaaS, AI agencies, and technical hiring**.
- Became the **only intern promoted to Senior Software Engineering Intern after two months**; mentor junior interns in programming, problem solving, delegation, communication, marketing, and outreach strategy.
- Currently optimize the UI of **applisolutions.com** and convert the web product into a dedicated **Android application** to expand mobile reach and accessibility.

Technologies: Python, NLP, RAG, LLM Fine-Tuning, AI-Powered Tools, Git, Web Scraping, Data Processing, Software Debugging, Android Development

Robotics Team Lead | STEAM Robotics

Sep 2024 – Present

- Led development of a competition rover across **embedded C++ firmware, circuit integration, sensors, mechanical systems, and hardware/software interfaces**.
- Implemented and iteratively tested Arduino/C++ control logic while integrating physical hardware with software behavior across multiple rover subsystems.
- Diagnosed **hardware-firmware synchronization failures** using rapid test cycles, live fault isolation, and systematic debugging across electrical and software boundaries.
- Reorganized development into parallel mechanical, circuit, and embedded workstreams when build time was reduced to four one-hour sessions, then delivered a fully functioning rover and led the team to **1st place**.

Technologies: C++, Arduino Microcontrollers, Embedded Firmware, Hardware/Software Co-Design, Circuit Integration, Hardware Diagnostics, Fault Isolation, Iterative Testing

Independent Embedded Systems Designer | Home Electronics Lab

Sep 2023 – Present

- Designed and built multiple embedded systems, including a **distance-sensing radar, automated trash detection mechanism, and soil-moisture monitoring system**.
- Developed Arduino/C++ firmware integrating microcontrollers, sensors, circuits, software interfaces, and online visualization across the physical-digital boundary.
- Calibrated sensors and debugged non-responsive components through repeated test cycles, isolating failures across wiring, circuits, firmware, and interfaces.
- Applied **power management, signal processing, circuit design, and sensor integration** to improve system reliability and embedded performance.

Technologies: C++, Arduino Microcontrollers, Embedded Systems, Sensor Integration, Circuit Design, Power Management, Signal Processing, Hardware Debugging

AWARDS & ACHIEVEMENTS

CSMC: School Champion, Top 5.05% | **Fermat:** School Champion, Top 5.75% | **Euclid:** School Champion, Top 13.23%

COMC: School Champion, Top 25% | **Schulich Leader Nominee:** 1 / 2,300 | **AP Scholar** | **Honor Roll:** 4 years